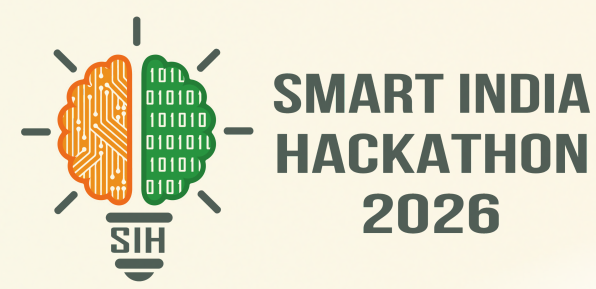




SMART INDIA HACKATHON 2026



- **Project Title:** Smart Municipal Waste Collection & Resource Optimization
- **Project Description:** Predict waste accumulation and optimize collection routes, vehicle deployment and intervention priorities for municipal authorities
- **PS Category:** Software

Team Name: HACK404



The Problem

Reactive Waste Management Leads to Overflowing Public Spaces

Current waste management strategies are **too reactive**, resulting in overflowing bins that create unsightly and unsanitary conditions in public areas, impacting community health and aesthetics.

NAGARAI'S PREDICTIVE APPROACH

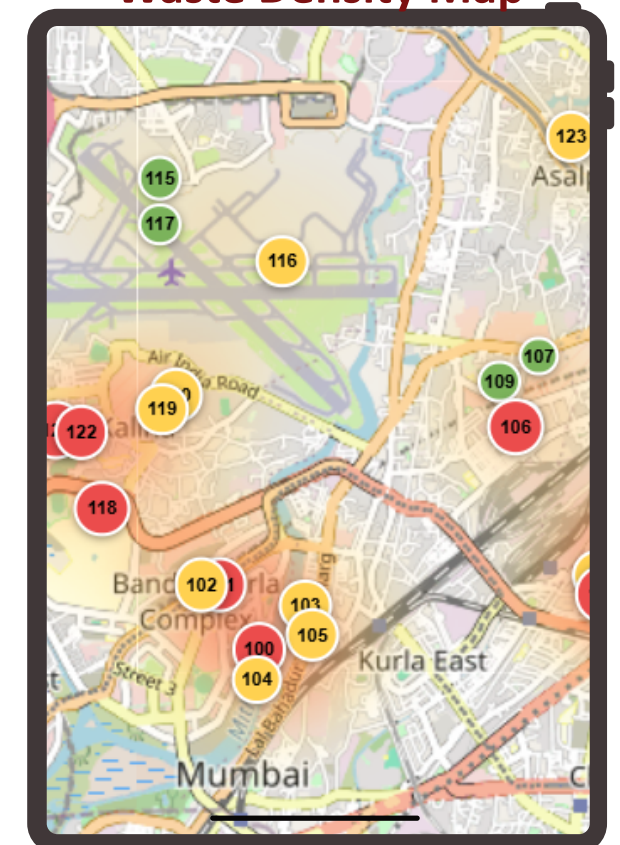
NagarAI leverages **advanced forecasting** techniques to predict waste accumulation early, enabling proactive collection, preventing overflow, and optimizing routes.

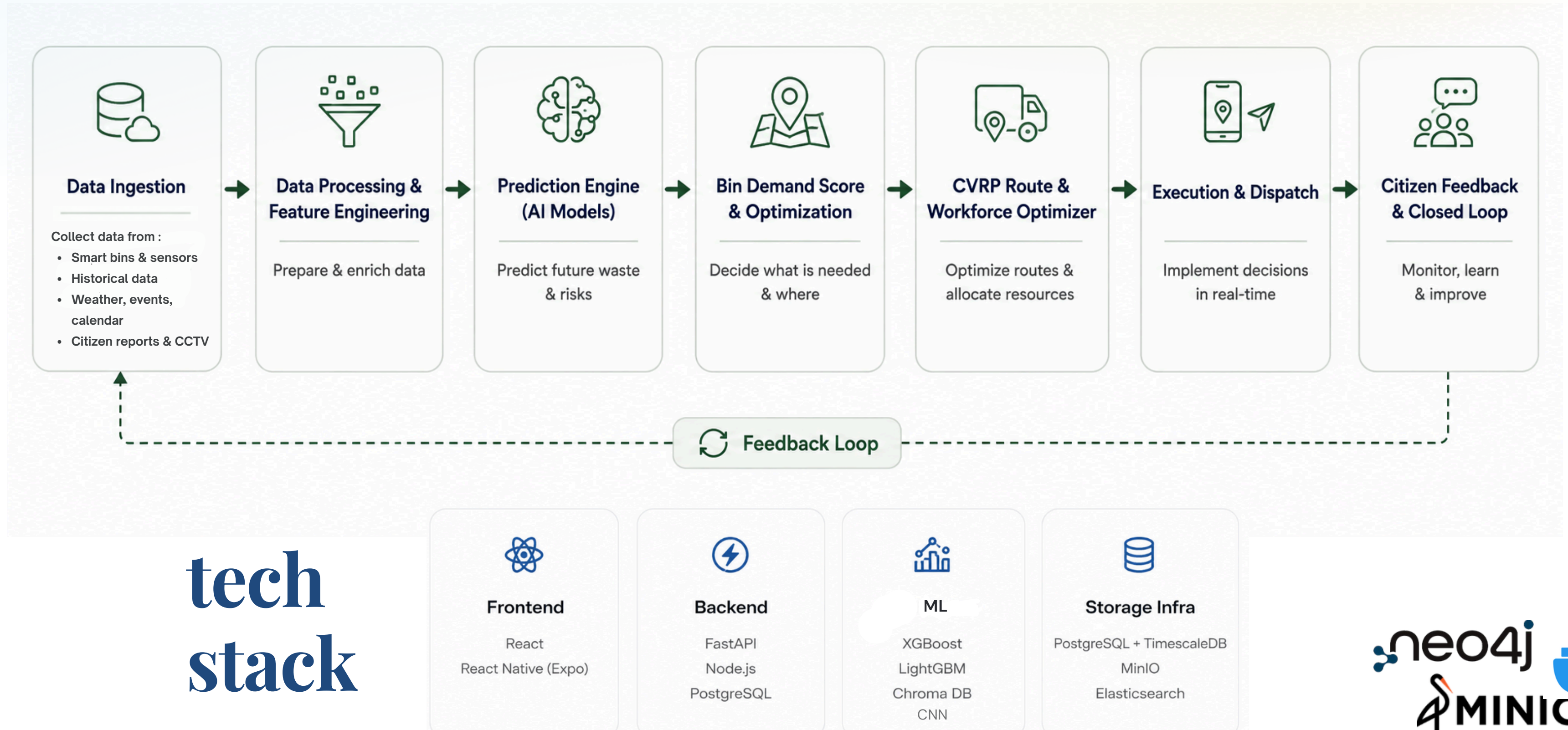


Innovation

Unlike bin-monitoring apps, NagarAI's **AI Intervention Engine** decides the right action for every hotspot – collect, relocate, resize, or run an awareness drive – not just that a bin is full.

Waste Density Map





PROVEN STACK

NagarAI utilizes a **proven MERN stack** along with standard ML libraries, ensuring no exotic infrastructure is required and avoiding vendor lock-in for seamless deployment.

MODULAR ROLLOUT

The solution allows for a **modular rollout**, starting with a single ward or module, enabling scalable implementation across the entire city progressively as needed.

MUNICIPAL DATA SILOS

Disparate municipal systems make data integration and sharing challenging.

CCTV PRIVACY

Privacy concerns require anonymizing surveillance data for safe use.

WORKER ADOPTION

Employees may resist new systems, so workflows must stay simple.



**potential
risks**



CITIZENS

Citizens benefit from **cleaner public spaces** through efficient waste management and reduced overflow.



MUNICIPAL WORKERS

Municipal workers receive **prioritized tasks**, improving productivity and reducing guesswork.



OFFICIALS

Officials gain **evidence-based** insights for better planning, preparedness, and resource allocation.



Social impact

Higher civic trust through cleaner public spaces and fewer overflowing bins.



Economic advantage

Lower operational costs via optimized routes and reduced fuel use.



Environmental benefit

Reduced CO2 emissions and better waste segregation practices.



Governance improvement

Data-driven planning for better infrastructure investment.

Upto **36%**
Collection Distance Reduction

NagarAI's innovative approach aims to **reduce operational costs** by optimizing waste collection routes & minimizing unnecessary trips, cutting travel distance by 36.8% and saving 28.2% time. [\[1\]](#)

[1] AI for Smart City Waste Management

Anthes, R. et al. (2023). Artificial intelligence for waste management in smart cities: a review. Environmental Chemistry Letters.

AI-driven waste logistics reduce transport distance by up to 36.8%, costs by 13.35%, and time by 28.2% — the basis for NagarAI's collection-distance reduction target.

[2] ML Forecasting of MSW Generation

Solid Waste Generation and Disposal Using Machine Learning Approaches: A Survey of Solutions and Challenges (2022). MDPI Sustainability.

A gradient-boosting regression model achieved 88% accuracy forecasting short-term waste generation in New York City — validating NagarAI's use of XGBoost/LightGBM.

[3] Smart Bin Case Study – Melbourne

Graph-based predictive modeling for waste management in smart cities. Neural Computing and Applications.

LSTM/GRU models predicted bin fill levels using real sensor data from a Melbourne-area municipal deployment – a working precedent for NagarAI's Live Waste Pressure Score.